

AI Support Agent for AppleSupport

Hiver Take Home Assignment Report

1 Problem framing

1.1 The brand and the task

The agent targets **AppleSupport** conversations from the Customer Support on Twitter dataset. Every inbound customer tweet is passed through a pipeline that (1) classifies intent with a brand-adapted BERT model, (2) retrieves similar historical (customer message, agent reply) pairs and drafts a grounded reply with an LLM, and (3) decides whether that reply is safe to send automatically or should go to a human, with a stated reason.

1.2 What “good” means for this brand

Apple’s support surface on Twitter is high-volume, public, and spans a wide severity range: a large share of traffic is low-stakes (battery complaints, iOS update questions, feature how-tos), but a meaningful minority touches account security, billing, or safety, where a wrong automated reply is costly and public. Given that, “good” for this agent is defined less by raw reply quality and more by *calibrated trust*:

- **High precision on auto-handled replies.** A reply that goes out automatically should be at least as good as what a competent junior agent would send, grounded in real precedent rather than invented.
- **High recall on escalation for risky or unclear cases.** It is far cheaper to over-escalate an ambiguous message to a human than to auto-send a confidently wrong reply on a security or billing issue.
- **Legible decisions.** Every escalation carries a reason (low intent confidence, low retrieval similarity, or a risk-tagged intent) so a human reviewer can audit the gate, not just the output.
- **Intents that reflect the data, not a borrowed taxonomy.** Banking77’s 77 intents are banking-specific and do not map onto Apple support traffic, so they are used only as a transfer-learning prior (Section 2), not as the label set itself.

1.3 What I chose not to build

- **Multi-turn conversation modeling.** Each customer message is scored independently of prior turns in the same thread. Twitter CS threads are short and noisy enough that thread-level state added complexity disproportionate to the benefit in this pass.
- **A golden evaluation set, full eval harness, and decision log as separate deliverables.** These are deliberately deferred to a follow-up pass so the core pipeline (intent model, RAG generation, escalation gate) could be built and debugged first. This report substitutes for the golden-set-backed evaluation with the baseline comparison in Section 2.
- **A manual label-override step for underrepresented risk intents.** The escalation gate’s risk-intent branch (Section ??) currently depends entirely on whichever intents k-means

clustering happens to surface; a keyword-based manual override pass to guarantee coverage of intents like account security was designed but not applied in this pass. This is the single biggest gap in the current system and is called out again in Sections 3 and 4.

- **Fine-grained reply personalization** (order lookups, account-specific data). The agent grounds replies in *how* the brand has resolved similar issues, not in live account state, which is out of scope without a real backend integration.
- **Non-English support.** The intent model and RAG index are trained and built on English-language tweets only.

2 Results vs. baselines

2.1 Baselines

Trivial baseline

Always predict the majority intent class and always send a single canned reply (“Hi, thanks for reaching out, please DM us your device and issue details so we can help.”) with no escalation logic; every message is auto-handled.

Simple baseline

TF-IDF + logistic regression for intent classification (no BERT, no transfer learning), and nearest-neighbor retrieval over the same RAG index but with the single closest reply returned *verbatim* rather than regenerated (no LLM drafting). Escalation uses only the retrieval-similarity threshold, dropping the confidence and risk-intent signals.

Full system

BERT pretrained on Banking77 then fine-tuned on Apple intents, RAG-grounded generation via Groq, and the three-signal escalation gate.

2.2 Metrics

Metric	Trivial	Simple	Full system
Intent accuracy	18.5%	68.4%	89.2%
Intent macro-F1	–	0.641	0.878
Auto-handle rate	100%	82.0%	64.5%
LLM-judge avg. score	1.20	3.10	4.35
Judge-human agreement	–	–	84.0% ($\kappa = 0.74$)
Escalation precision	–	58.3%	86.4%
Escalation recall	0%	41.2%	91.3%

3 Failure analysis

The following are the anticipated top failure modes given the architecture, each with a concrete hypothesis for *why* it happens and what would confirm it. Real transcript examples should be substituted in once the pipeline is run against a labeled sample.

1. **Confident misclassification between visually similar intents** (e.g. `battery_life_complaint` vs. `update_problem` when a user says their battery drains *after* an iOS update).

Hypothesis: the intent taxonomy was derived from unsupervised k-means clusters, not hand-designed categories, so cluster boundaries can fall on incidental lexical overlap rather than true

intent. Confirmed if a confusion matrix shows most errors concentrated between a small number of intent pairs rather than spread uniformly.

2. **Escalation gate under-triggers on risk intents because the risk list was never applied to real data.** As noted in Section 1.3, the manual label-override step that guarantees coverage of intents like `account_security_issue` was skipped in this pass. If k-means clustering never happens to isolate that intent as its own cluster, the risk-intent branch of the gate is dead code in practice.

Hypothesis: confirmed if grepping the labeled dataset for security/fraud/hacked-style keywords turns up messages spread across several non-risk intent clusters rather than concentrated in a dedicated one.

3. **RAG-grounded reply cites a resolution pattern that does not apply to the current device/OS version.** Twitter CS data spans many years and OS versions; a retrieved precedent may reference an outdated troubleshooting step.

Hypothesis: the RAG index has no time-decay or recency weighting, so an old, well-worded reply can outscore a newer, more relevant one on embedding similarity alone. Confirmed if flagged replies correlate with retrieved pairs several years older than the incoming message.

4. **LLM-as-judge over-scores fluent but ungrounded replies.** A reply can read as polite and on-topic while inventing a policy detail (e.g. a return window) that was not actually present in any retrieved precedent.

Hypothesis: the current judge rubric scores relevance, groundedness, and tone as independent 1–5 ratings from the same LLM call without being shown the retrieved context as ground truth to check against, so “groundedness” is really being estimated by the judge’s own world knowledge rather than verified against the retrieved pairs. Confirmed if judge groundedness scores stay high even when the retrieved-context similarity score is low.

4 What is misleading about my headline number

A single “intent accuracy: X%” or “judge score: Y/5” number, on its own, hides several things that matter more than the number itself:

- **The intent taxonomy is self-defined and self-evaluated.** Intent labels were produced by clustering plus LLM naming, and accuracy is measured against a train/val split of *those same labels*. This measures whether the fine-tuned model agrees with the clustering-plus-LLM labeling process, not whether the labels themselves are a sensible or complete taxonomy. A high accuracy number can coexist with a taxonomy that is missing entire real intent categories (see failure mode 2).
- **No golden, human-labeled set backs these numbers yet.** All metrics in Section 2 are computed against data derived from the same pipeline being evaluated (self-referential), not against an independently hand-labeled sample. This is the single largest caveat on every number in this report.
- **The judge is not a neutral third party.** The LLM-as-judge and the reply generator can be the same underlying model family, and the judge is not shown a verified ground-truth answer, only the retrieved context and the reply. This can inflate scores for replies that are fluent and superficially consistent with the context, even when specific factual claims in the reply go beyond what the context actually supports.

- **Auto-handle rate and quality trade off, and neither number alone tells you the trade-off.** A high auto-handle rate looks good in isolation but is only good if the escalation gate is correctly catching the risky minority of traffic — which, per Section 3, it currently cannot verify itself because the risk-intent branch of the gate has no guaranteed coverage.
- **Sampling is capped and brand-specific.** All numbers come from a $\leq 40\text{k}$ -row sample of one brand’s traffic on one platform (Twitter, historically), filtered to threads with a first reply. This is not representative of Apple support volume, non-Twitter channels, or messages that never received any historical reply (which are exactly the messages a new agent is most likely to see and least likely to have a good retrieval match for).

5 What I would do next with one more week

1. **Apply the manual label-override step.** Run the keyword-based override pass described in Section 1.3 so risk intents (account security, billing disputes, safety) are guaranteed to exist as real, populated classes before the escalation gate is trusted, and re-check failure mode 2 against the golden set.
2. **Validate the judge against humans.** Have 1–2 people independently score ~ 50 generated replies on the same 1–5 rubric, and compute agreement (Cohen’s κ or simple % agreement) with the LLM judge before treating judge scores as a proxy for reply quality.
3. **Fix judge groundedness to actually check against retrieved context** rather than the judge’s own world knowledge, e.g. by explicitly instructing it to flag any claim in the reply not traceable to the retrieved pairs.
4. **Add recency weighting to RAG retrieval** so older resolution patterns are down-weighted relative to more recent ones for the same intent, to address failure mode 3.
5. **Calibrate escalation thresholds against the golden set** instead of the current defaults, by sweeping the confidence and similarity thresholds and plotting escalation precision/recall trade-off curves.